

Carlo Squarcia Mateo

Industrial Informatics and Robotics Engineer | Upcoming Master's in Artificial Intelligence for Industrial and Manufacturing Processes

Valencia, Spain | +34 603 647 762 | [Email](#) | [LinkedIn](#) | [GitHub](#)



Professional Summary

Graduate in **Industrial Informatics and Robotics** from the **Universitat Politècnica de València (UPV)**, with one year of international academic experience at **Politecnico di Torino**, where I completed specialized courses in **artificial intelligence, cybersecurity and computer vision**.

Starting a **Master's Degree in Artificial Intelligence for Industrial and Manufacturing Processes** at **UNIR** in November 2026.

Currently completing a structured advanced ROS 2 training path, covering Python and C++ development, TF transformations, URDF robot modeling, Gazebo simulation, autonomous navigation, `ros2_control`, manipulation and robotic perception.

Specialized in **industrial process and robotic cell simulation** using tools such as **RoboDK** and **FlexSim**, as well as in **applied artificial intelligence**, including convolutional, recurrent and bidirectional neural networks.

Strong background in **2D and 3D computer vision**, image processing, object classification and scene analysis using **Python, OpenCV** and **deep learning techniques**.

Technical Skills

Robotics and Simulation: ROS 2, Python/C++ ROS 2 development, nodes, topics, services and custom interfaces, TF2, URDF, Gazebo, Nav2, `ros2_control`, MoveIt, robotic manipulation and perception, RoboDK, FlexSim, path planning and digital twins.

Artificial Intelligence: Machine Learning, Deep Learning, neural networks (CNN, RNN and BiRNN), EfficientNet, Natural Language Processing (NLP), Transformers, BERT, multiclass classification, TensorFlow and PyTorch.

Computer Vision: 2D and 3D computer vision, OpenCV, image processing, segmentation, object detection and classification, 3D reconstruction and camera calibration.

Programming and Tools: Python, C/C++, SQL, Git, Linux and MATLAB/Simulink.

Industrial Automation: PLCs, Ladder Logic, GRAFCET/SFC, process control, embedded systems, microcontrollers, Industry 4.0 and IIoT.

Cybersecurity and Communications: industrial cybersecurity and industrial networks.

Selected Projects

Intelligent Robotic Assembly System with Computer Vision and Deep Learning

RoboDK · UR5e · OpenCV · EfficientNet-B0 · [GitHub](#)

Developed an autonomous robotic assembly cell in **RoboDK** using two **UR5e robots**, integrating **OpenCV** and **EfficientNet-B0** to detect, classify and estimate the position of parts on a conveyor belt. Achieved **97.92% classification accuracy** and implemented grasping logic, image-to-robot coordinate transformation through homography, and dynamic precedence management to coordinate **four assembly stations**, validating the system in continuous operation with **12 completed assemblies**.

Sim-to-Sim Transfer Using Reinforcement Learning

Python · MuJoCo · PPO · Domain Randomization · PCA · t-SNE · [GitHub](#)

Trained robotic locomotion policies using **PPO in MuJoCo**, applying **Domain Randomization** to physical parameters to improve generalization across simulation environments. Increased the return in the target environment from **835 to 1320 (+58%)** and achieved a **100% episode completion rate without falls**, while analyzing robustness using **PCA** and **t-SNE**.

Driver Assistance Systems Using Computer Vision

Python · OpenCV · Computer Vision · Driver Monitoring · [GitHub](#)

Developed **Python-based** lane detection and driver monitoring systems, processing bird's-eye-view images to identify road lane markings and analyzing **head orientation, eye closure and heart rate** in real time to detect **driver distraction, microsleep and drowsiness**.

Automated Baggage Handling Using AGVs

FlexSim · AGVs · A Path Planning · Industrial Automation

Designed and simulated in **FlexSim** an automated airport baggage classification and transportation system using **AGVs**, conveyor belts and sensors. Implemented vehicle assignment logic, baggage grouping by identifiers and *A route planning** for standard and premium services.

Cyberattack Tactic Detection and Classification Using Transformers

Python · BERT · UniXcoder · Transformers · MITRE ATT&CK · [GitHub](#)

Developed a command classification system for **SSH sessions** to identify attack tactics based on **MITRE ATT&CK**, comparing Transformer models **BERT** and **UniXcoder**. Implemented preprocessing and token-level label alignment, fine-tuning, hyperparameter optimization and layer-freezing strategies, achieving **86.7% accuracy** and a **macro F1 score of 0.724** with UniXcoder on the test set.

Education

Master's Degree in Artificial Intelligence for Industrial and Manufacturing Processes

UNIR · Online · Starting November 2026

Bachelor's Degree in Industrial Informatics and Robotics Engineering

Universitat Politècnica de València (UPV) · Valencia, Spain · September 2021 – September 2026

International Academic Exchange — Politecnico di Torino

Politecnico di Torino · Turin, Italy · September 2025 – June 2026

Completed a full academic year at Politecnico di Torino as part of the UPV exchange program, taking specialized courses in **artificial intelligence**, **cybersecurity and computer vision** not included in my home degree curriculum and gaining experience in an international academic environment.

Certifications

- **GitHub Foundations Certification** — GitHub · 2026 · [View credential](#)
- **Advanced ROS 2 Learning Path — The Construct** · In progress · 2026 Structured hands-on training covering ROS 2 development in Python and C++, URDF, TF2, Gazebo, Navigation and Advanced Navigation, ros2_control, manipulation, perception, and development of a complete ROS 2-based robot.

Languages

- **Spanish:** Native
 - **Valencian:** C1
 - **English:** B2
 - **Italian:** Basic
-